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Interactive Computer Graphics

Chapter 8 – Local Illumination

Summer Term 2026

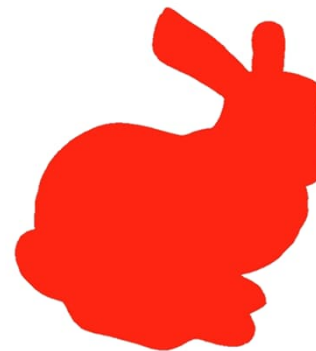
Timo Ropinski



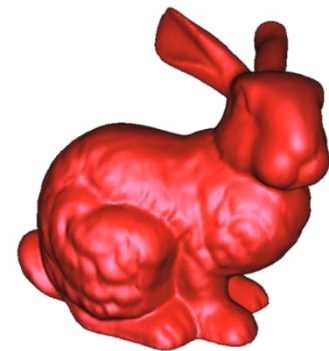
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Local Illumination

- Illumination is the process of simulating light interactions in a virtual scene
 - Requires light and material properties
 - Achieved through illumination model
 - Essential for scene perception



without illumination



with illumination

- Two types of illumination are differentiated
 - Local illumination – considers only light directly reaching scene object
 - Global illumination – also considers light modified by other scene objects (e.g., shadows, reflections)

Chapter Outline

8.1 Physics of Light

8.2 Light Sources

8.3 Material Models

8.4 Phong Illumination Model

8.5 Shading

References



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8.1 Physics of Light

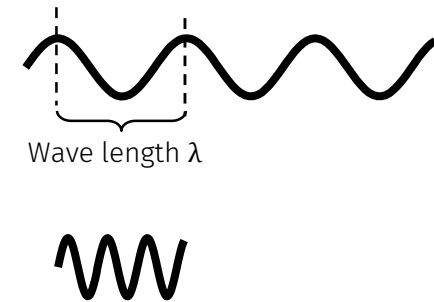
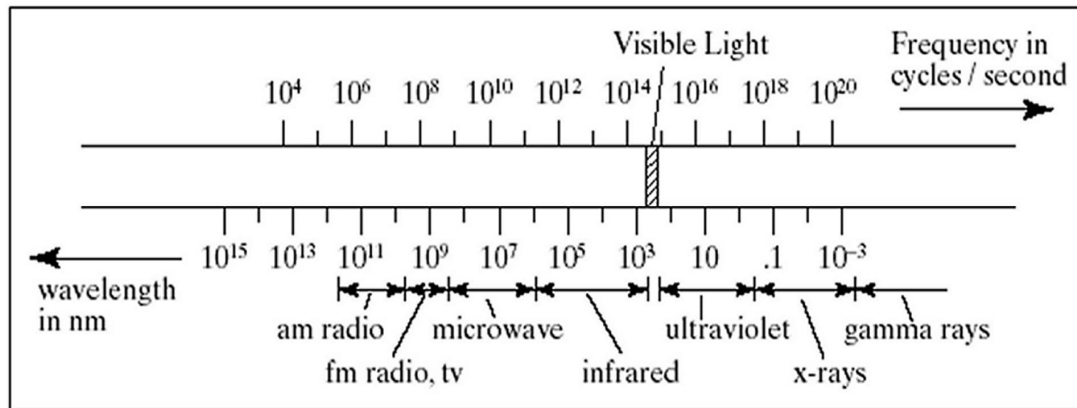
Understanding light in the real world



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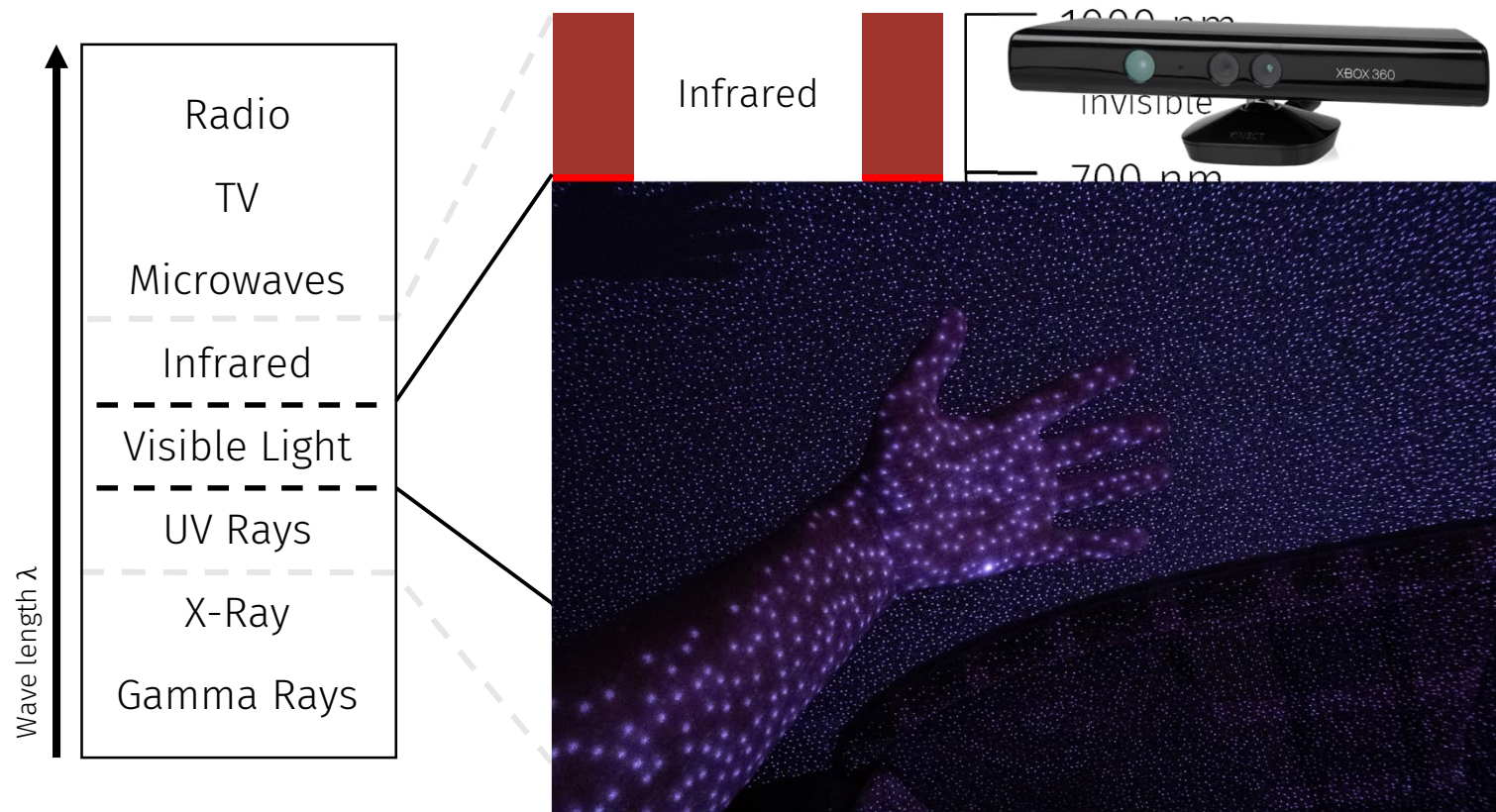
Electromagnetic Radiation

- Electromagnetic radiation is defined by a spectrum of wavelengths

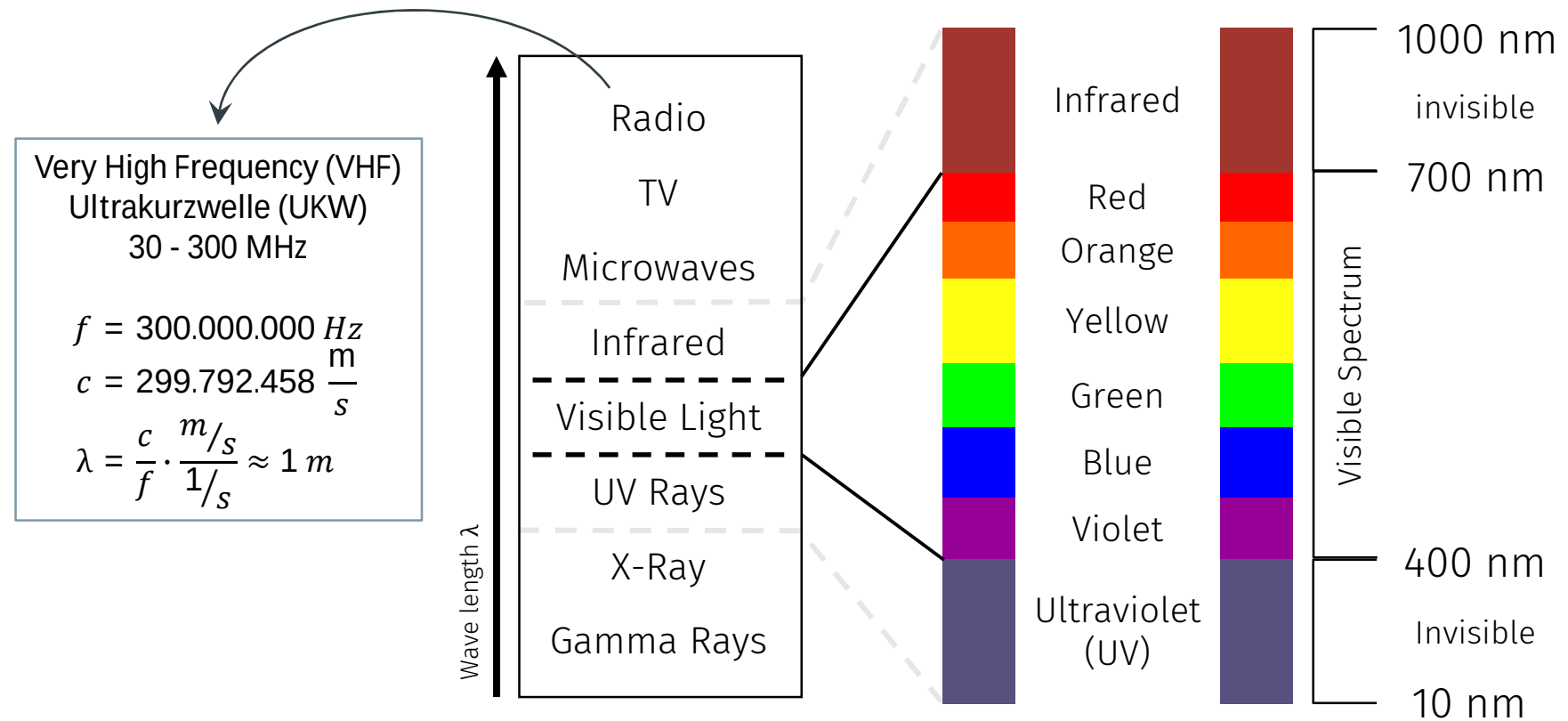


- Visible light is subset of electromagnetic spectrum between 400 and 700nm, which humans perceive as colors

Visible Light

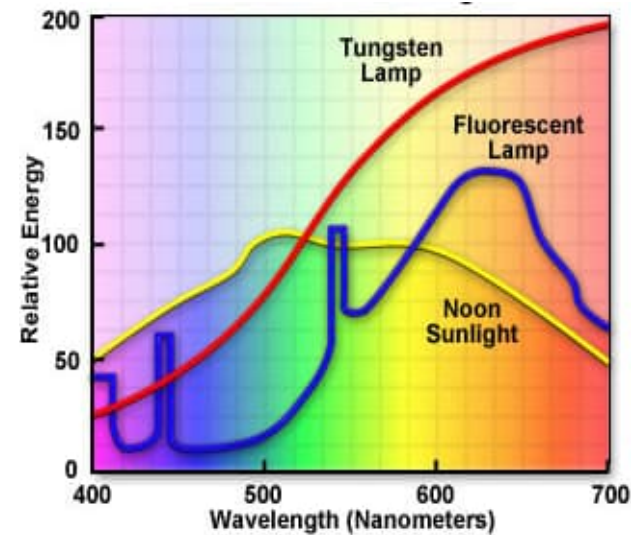
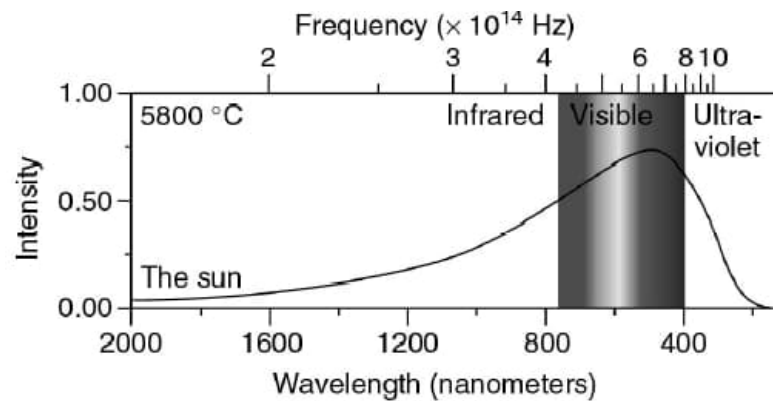


Visible Light



Light Spectra

- Different light sources have different light spectra
 - Intensity differs based on wavelength in visible spectrum

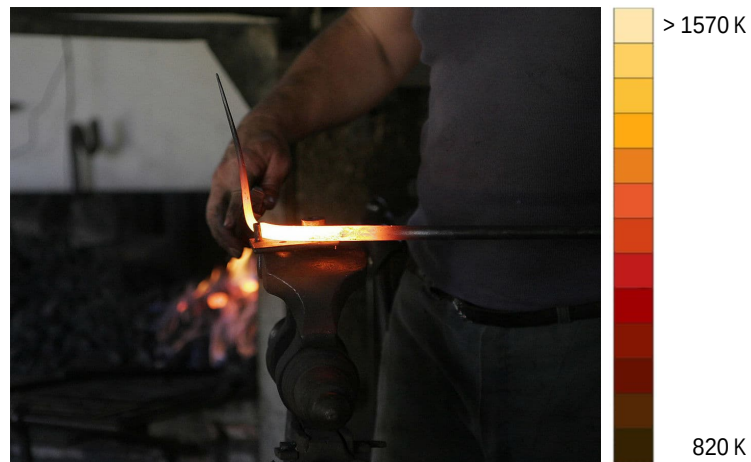


[<http://www.micro.magnet.fsu.edu/optics/lightandcolor/sources.html>]

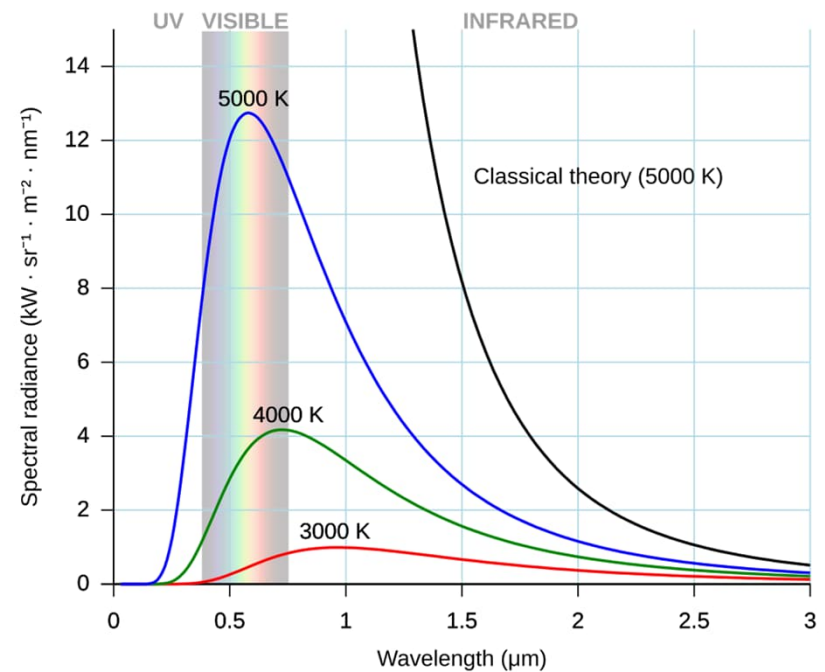
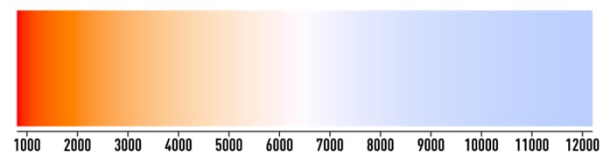
Black-Body Radiation

- A continuous spectrum emitted by a black body at a certain temperature

Steel



Stars

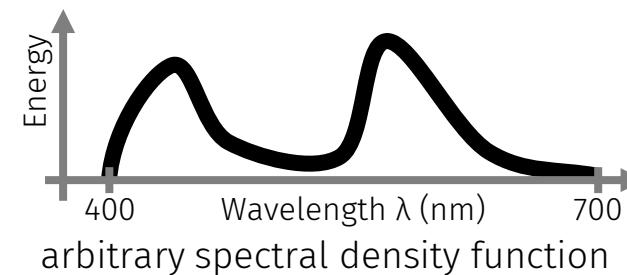
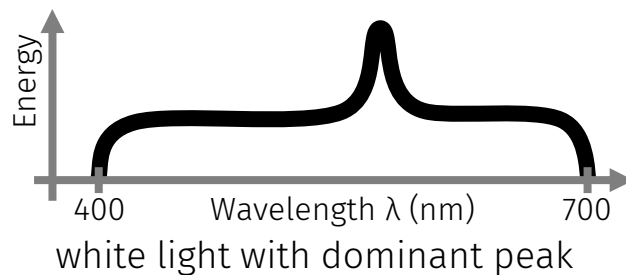
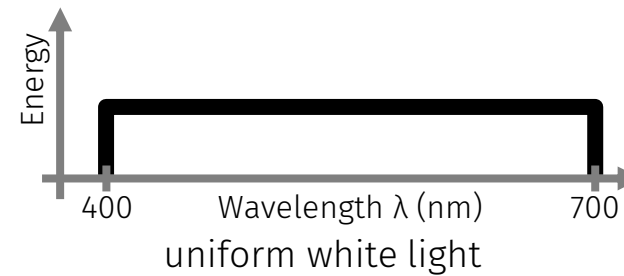
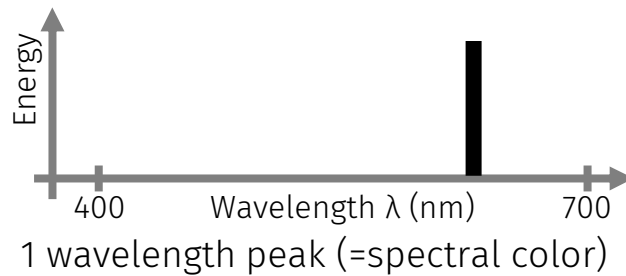


Black-Body Radiation



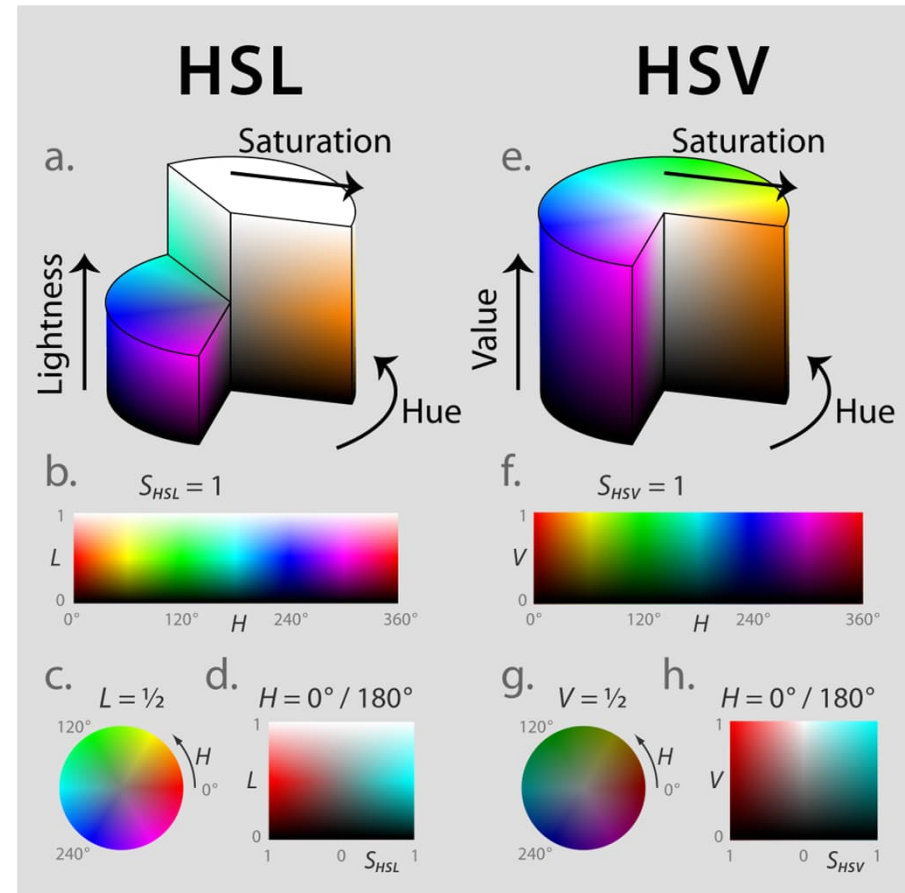
Spectral Density Function (SDF)

- Light spectra are represented as spectral density functions
- Spectral density functions provide the light intensity (=energy) per wavelength



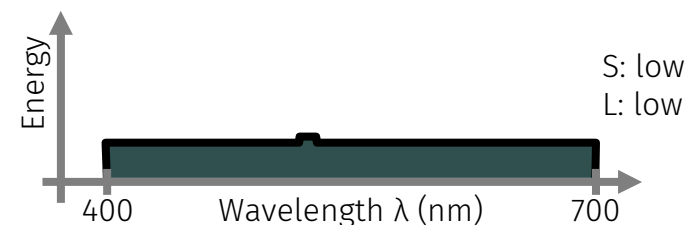
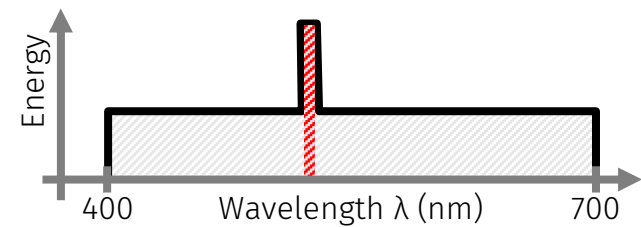
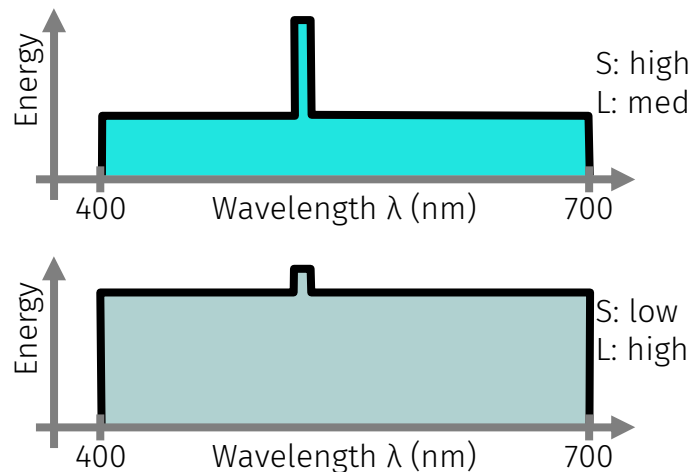
Light Perception

- Humans perceive light spectra as differently colored lights
- Often described using three main properties (HSL/HSV model)
 - Hue - primary color tone
 - Saturation - color intensity
 - Lightness/Value – brightness
- Also see [Matplotlib Colormaps](#)



Interpreting SDFs

- Shape of an SDF depicts how humans perceive represented light
 - Hue: dominant wavelength
 - Saturation: % of energy in dom. wavelength
 - Luminance: total energy (SDF integral)



Representing SDFs

- In spectral rendering SDFs are discretized as arrays
 - Cell number represents differentiable wavelength ranges
 - Cell values represent energy per wavelength range
- Much more frequently SDFs are represented as three-channel color
 - HSL values are derived from SDF
 - HSL values are converted to RGB values



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8.2 Light Sources

How to represent light sources in computer graphics



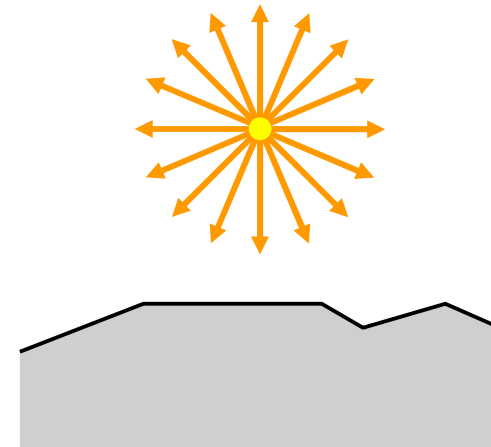
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Light Sources

- All light sources have a RGB color property representing their SDF
- Light sources are also defined through their location, orientation and extent within a virtual scene
 - Different light source types omit different spatial properties
- Light sources have no associated geometry
 - Do not have surfaces that reflect light (e.g., glass bulb for light bulb)
 - Are conceptual, but not geometric elements of a virtual scene

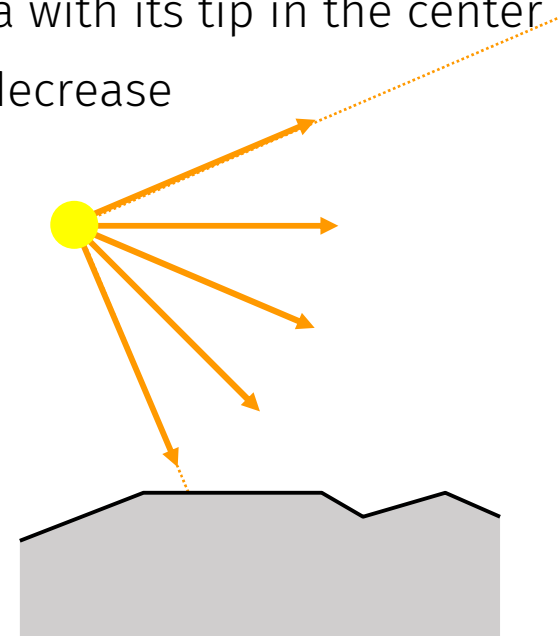
Point Light Sources

- Point light sources $L_p = (P, C)$ have a geometric center $P = (x, y, z)$ and radiate light with color $C = (R, G, B)$ evenly in all directions



Cone Light Sources

- Special point light source with geometric center
- Radiation of light restricted to a cone-shaped area with its tip in the center
- Specified by cut-off angle and exponents for the decrease of brightness with distance to the center (spot light exponent)



Light Attenuation

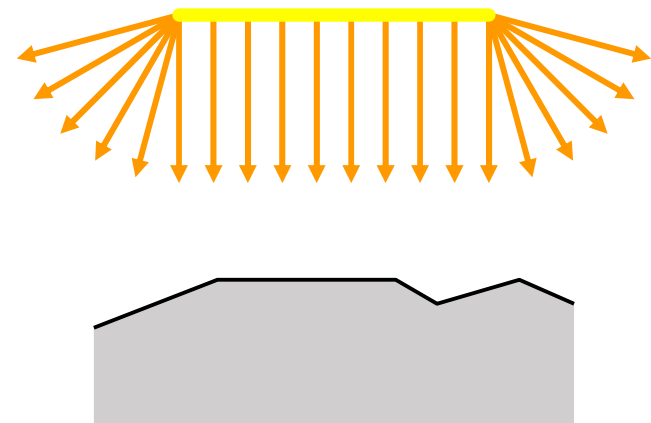
- Light attenuation is inversely proportional to the square of the distance d between the light source and the object
- Attenuation factor

$$f_{att} = \min\left(\frac{1}{c_1 + c_2 \cdot d + c_3 \cdot d^2}, 1\right)$$

- c_1 , c_2 and c_3 are user-defined constants associated with the individual light source
- By forming a minimum with **1**, the factor never increases the light intensity

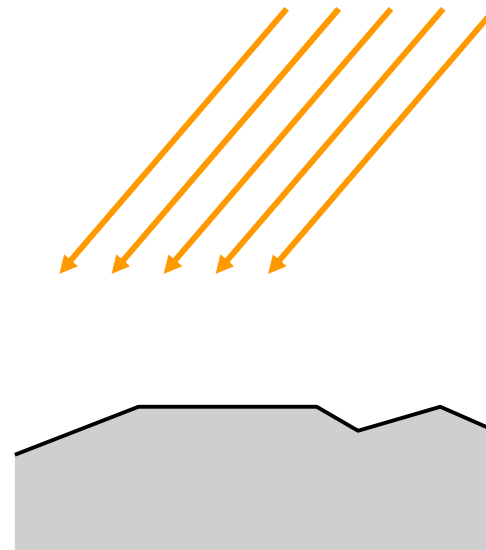
Area Light Sources

- Light source with geometric center and geometric extension
- Mostly of planar shape
- Transmission of light over the entire surface analogous to a point light source



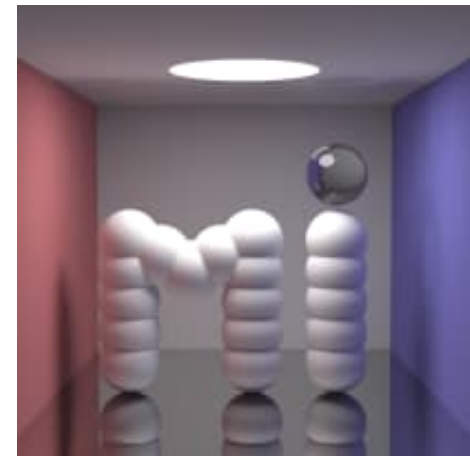
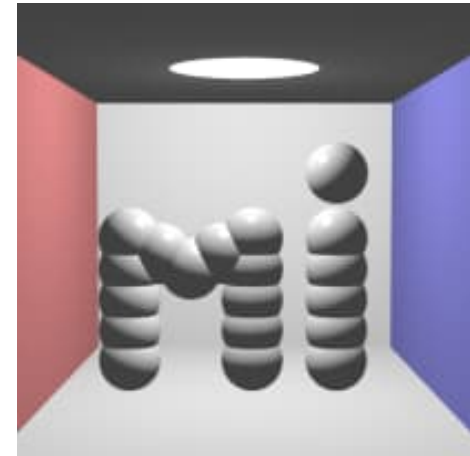
Directed Light

- Infinity-far away imaginary light source
- Light rays run parallel and evenly throughout the scene, direction is specified as a vector
- Light intensity does not diminish with increasing distance



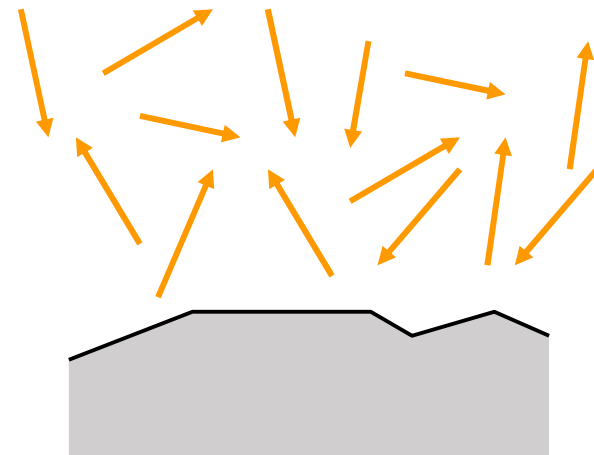
Local vs. Global Illumination

- Simple case: Local lighting
 - Consideration of direct illumination by light sources (without obstacles or reflection)
 - Simplified light models approximate real lighting effect
- More difficult case: Global lighting
 - Addition of indirect lighting (e.g., reflections / shadows between objects)
 - Complex equation systems depending on all scene objects are required



Ambient Light

- Undirected light rays, evenly distributed
- Light intensity constant, regardless of distance
- Simulates general room brightness (basic brightness)



8.3 Material Models

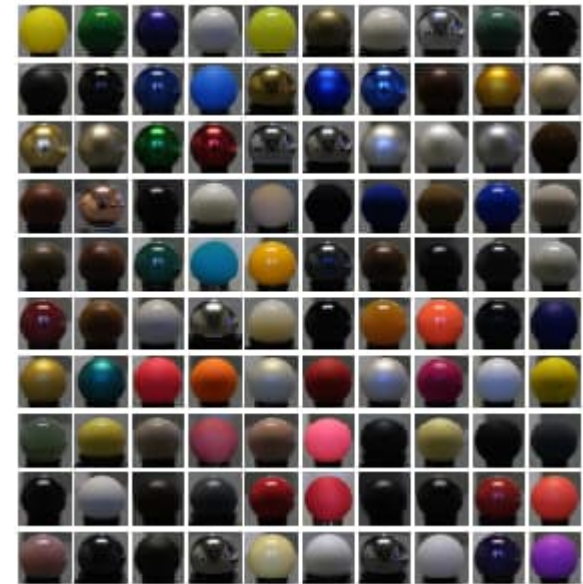
How to represent materials in computer graphics



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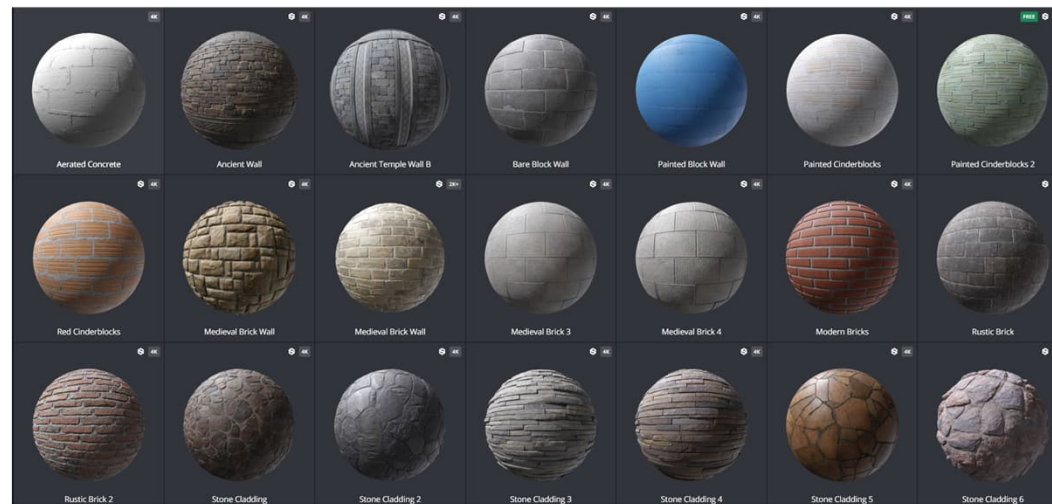
Material Models

- Material models capture the different properties of materials
- Materials can have a wide variety of appearances
- Different groups of materials can be classified based on appearance
 - Metal – shiny and specular appearance
 - Fabric – diffuse and soft appearance
 - Glass – translucent appearance
 - Wood – diffuse and structured appearance
 - ...



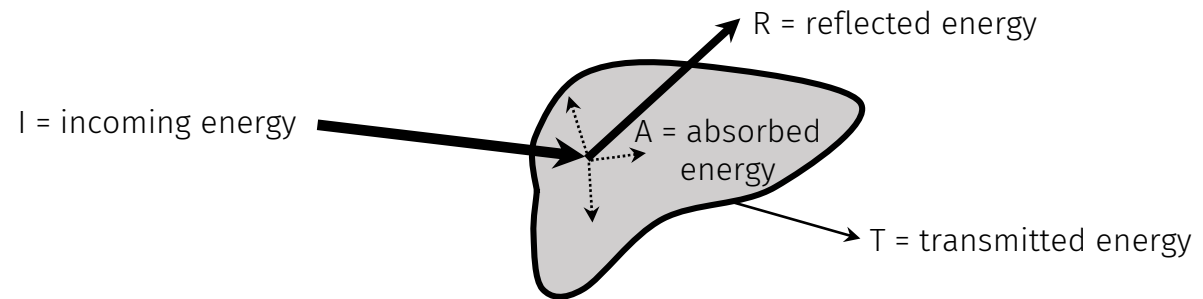
Materials vs. Textures

- Material models specify the material properties for an infinitesimal surface
 - No material variance with respect to location is supported (i.e., structures such as wood grain cannot be represented)
- Textures are able to support material variance with respect to location



Light Material Interaction 1/3

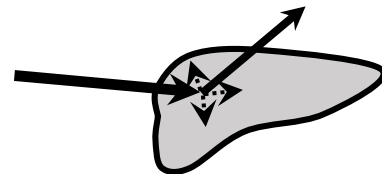
- Material appearance is influenced by three wavelength-dependent effects
 - Reflection – light energy bounces off
 - Absorption – light energy is reduced
 - Transmission – light passes through the object (refraction occurs)



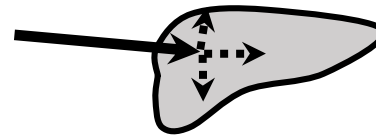
- Different materials showcase different combinations
 - Metal – reflection + absorption
 - Skin – reflection + transmission + absorption
 - Coal – absorption

Light Material Interaction 2/3

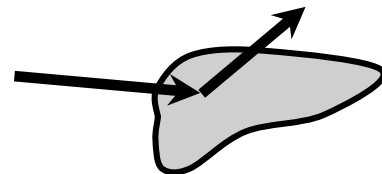
- Material properties determine weight of the individual effects



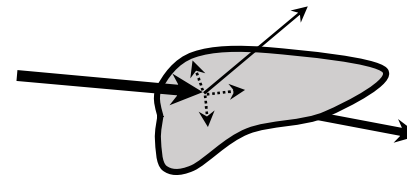
opaque colored material
 $A > 0, T = 0, R > 0$



opaque black material
 $A = 1, T = 0, R = 0$



perfect mirror material
 $A = 0, T = 0, R = 1$

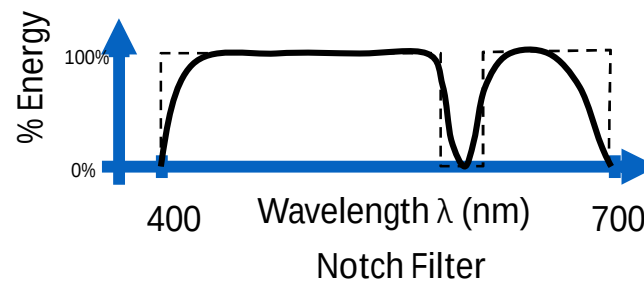
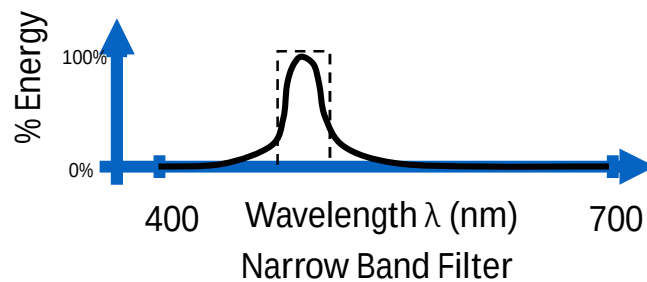
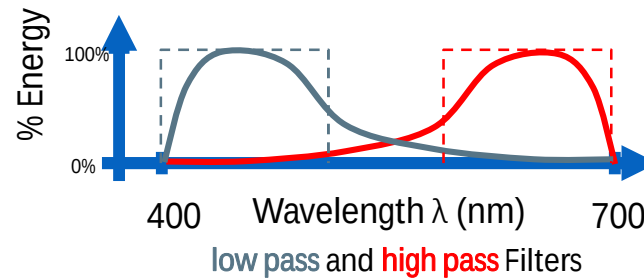
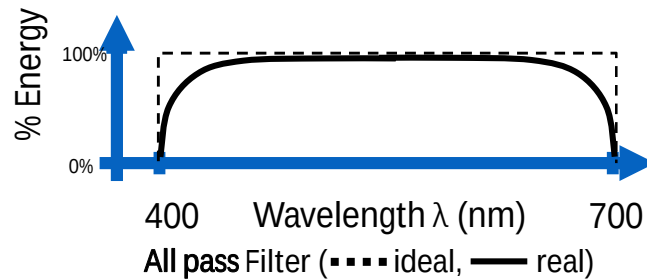


transparent material
 $A \approx 0, R \approx 0, T \approx 1$

- In this chapter we mainly consider absorption and reflection

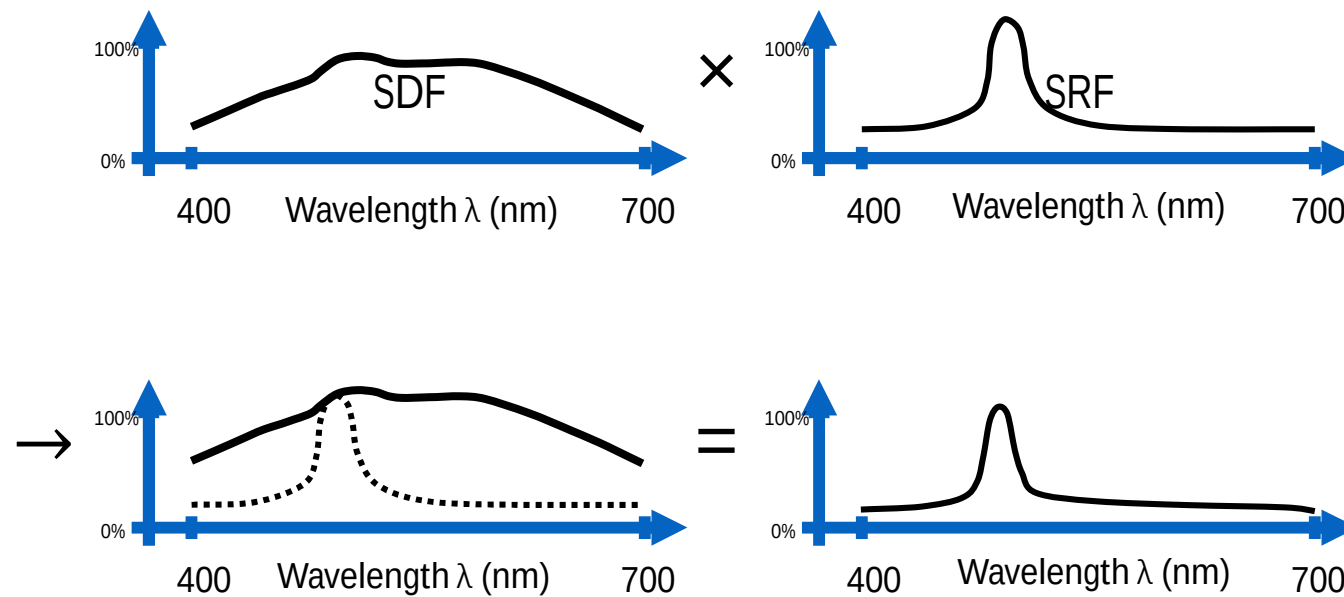
Light Material Interaction 3/3

- Light-material interaction described by the Spectral Response Function (SRF)



Reflection = SDF x SRF

- Light-material interaction results are the product of the SDF and SRF



Reflection Modeling 1/2

- All light sources can be described by their Spectral Density Functions
 - Natural light: sun, fire, ...
 - Artificial light: light bulb, laser, neon tube, LED, ...
- All material properties can be described by Spectral Response Functions
 - Glass, Water: SRF for transmission
 - Surfaces: SRF for reflection
- Sensors (camera pixels, film, eye) can be described as SRF
- Definition of visible color = integral over (result SDF x sensor SRF)

Reflection Modeling 2/2

- Synthetic and natural objects are in general not self-emitting
 - Light is reflected at its surfaces, i.e., re-emitted into the scene
 - Reflection is dependent on surface properties
 - Rough surface: scattering of light
 - Smooth surface: reflection of light
 - Different for individual wavelength ranges
 - Reflection depends on the angle of incidence of the light



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8.4 Phong Illumination Model

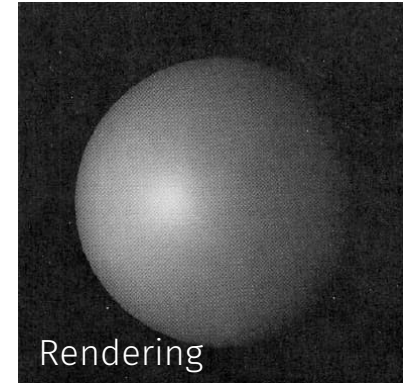
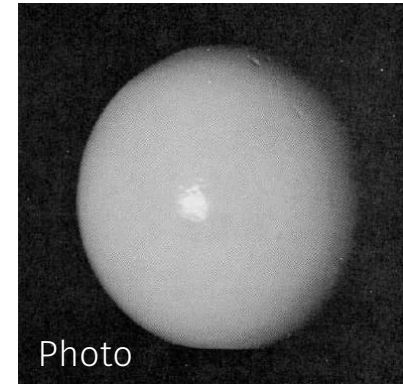
Determining the color of vertices



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Empirical Observations

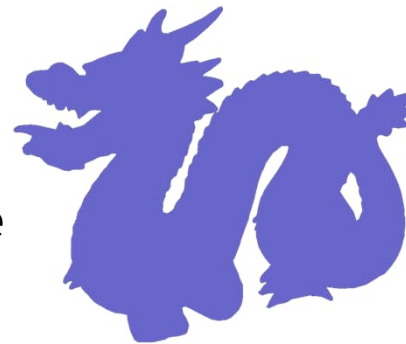
- Light reflections are
 - Dependent on the position of the observer
„... position of the observer...”
 - Dependent on the material properties of the surface
„... the specular properties of the object.”
 - Partly diffuse and partly specular
„...one part by the diffuse reflection and one part by the specular reflection...”



[Phong, 1975]

Illumination Models

- Illumination models calculate the color of scene objects based on the following properties
 - Location – Position $P = (x, y, z)$ of the selected point P
 - Orientation – normal direction $N = (n_x, n_y, n_z)$ of surface in P
 - Lighting – spatial properties and color properties of light source(s)
 - Material – material properties of the scene object
- Usually locations are given by fragment coordinates (image space)
- When computing illumination for vertices (object space) color needs to be spread over triangle surface ($\Rightarrow$ shading)



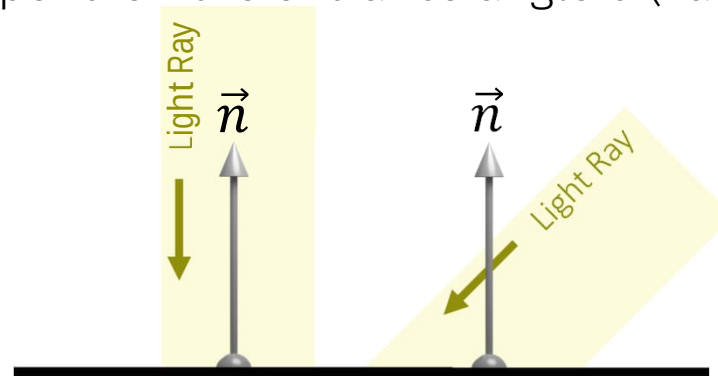
without illumination model



with illumination model

Diffuse Intensity 1/3

- Simulation of light reflections on diffuse surfaces
- Model assumption
 - Diffuse light reflections are non-directional (scattering)
- Observations
 - Diffuse light reflections are viewer-independent
 - Diffuse light reflections depend on the entrance angle θ (Lambert's law)



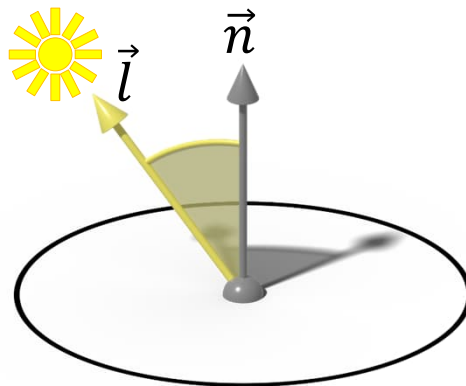
[Gouraud, 1971]

Diffuse Intensity 2/3

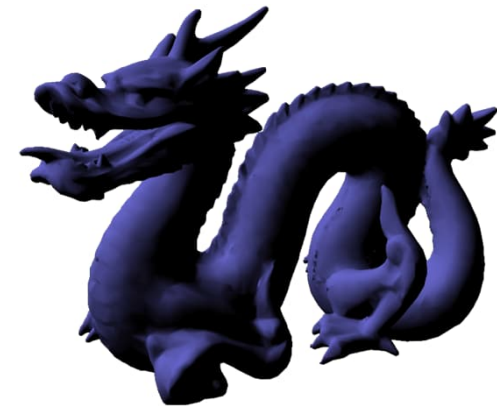
- Equation

$$I_{diffuse} = k_d \vec{n} \cdot \vec{L}_d \cdot \max(0, \quad)$$

- k_d diffuse material properties (surface dependent)
- L_d diffuse light intensity (light source dependent)



[Gouraud, 1971]



Diffuse Intensity 3/3

```
1 precision highp float;  
2 varying vec3 pos;  
3 varying vec3 n;  
4 varying vec3 l;  
5  
6 void main() {  
7     gl_FragColor = vec4(0.4, 0.4, 0.8, 1.0);  
8 }
```

Shdr



<https://bit.ly/2XFhbcM>

$$I_{diffuse} = k_d \cdot L_d \cdot \max(0, \vec{l} \cdot \vec{n})$$

Ambient Intensity 1/2

- Simulation of the general room brightness
 - Indirect light incidence or multiple reflections
- Model assumptions
 - Constant influence on all scene objects
 - No influence of position or alignment
- Equation
- $I_{ambient} = k_a \cdot L_a$
 - k_a ambient material properties (surface dependent)
 - L_a ambient light intensity (constant for scene)



Ambient Intensity 2/2

```
1 precision highp float;  
2 varying vec3 pos;  
3 varying vec3 n;  
4 varying vec3 l;  
5  
6 void main() {  
7     vec3 kd = vec3(0.4, 0.4, 0.8);  
8     vec3 Ld = vec3(1.0, 1.0, 1.0);  
9     vec3 I_diffuse = kd * Ld * max(0.0, dot(n,l));  
10  
11     gl_FragColor = vec4(I_diffuse, 1.0);  
12 }
```

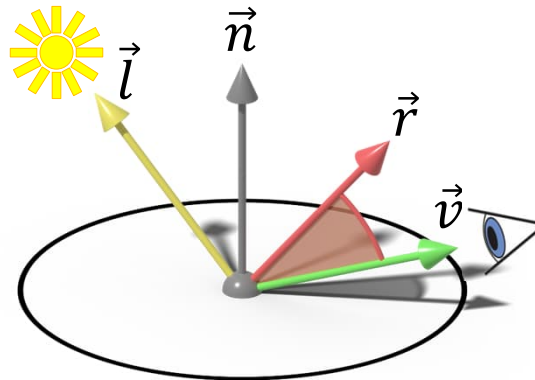


$$I_{ambient} = k_a \cdot L_a$$

<https://bit.ly/2XGn61n>

Specular Intensity 1/3

- Simulation of light reflections on specular surfaces
- Model assumption
 - Reflection of light are directed (reflection)
- Observations
 - Reflections of light are observer-dependent
 - Strongest reflection in mirror direction $\vec{r}$ and strong decrease of reflection by $\vec{r}$



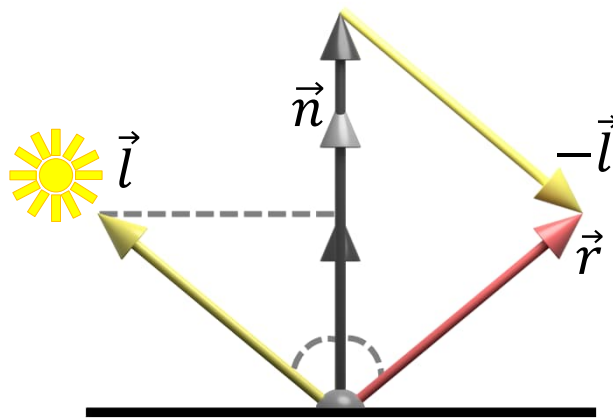
[Phong, 1975]

Reflection Vector $\vec{r}$

- Calculation of the reflection vector
 - Incidence angle corresponds to exit angle
- Equation

$$\vec{r} = 2 \cdot (\vec{n} \cdot \vec{l}) - \vec{l}$$

- $\vec{n}$ and $\vec{l}$ are normalized



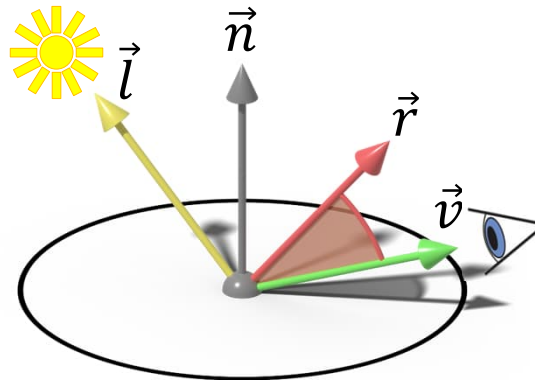
Specular Intensity 2/3

- Equation

$$I_{\text{specular}} = k_s \vec{v} \cdot L_s \cdot \max(0, \cos \theta)^p$$

- k_s specular material properties (surface dependant)
- L_s specular light intensity (light source dependent)

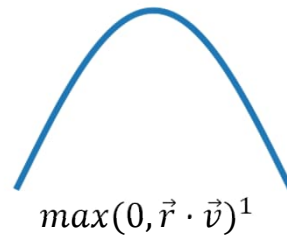
[Phong, 1975]



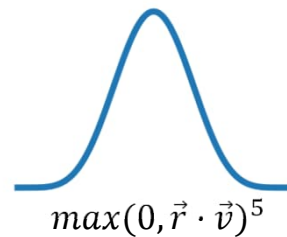
Specular Exponent p

- p influences fall off of specular highlight ($p \in [1, M]$ mit $M \approx 100$)

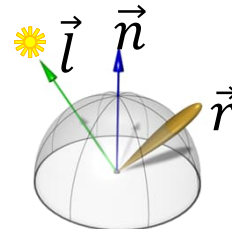
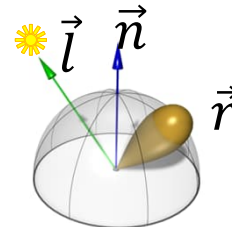
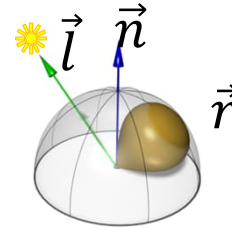
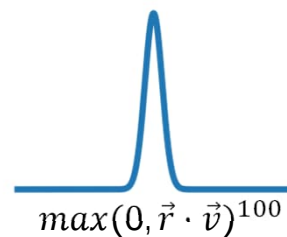
$p = 1$



$p = 5$



$p = 100$



[Phong, 1975]

Specular Intensity 3/3

```
1 precision highp float;  
2 varying vec3 pos;  
3 varying vec3 n;  
4 varying vec3 l;  
5  
6 void main() {  
7     vec3 kd = vec3(0.4, 0.4, 0.8);  
8     vec3 Ld = vec3(1.0, 1.0, 1.0);  
9     vec3 I_diffuse = kd * Ld * max(0.0, dot(n,l));  
10  
11     vec3 ka = vec3(0.3, 0.3, 0.3);  
12     vec3 La = vec3(0.5, 0.5, 0.5);  
13     vec3 I_ambient = ka * La;  
14  
15     gl_FragColor = vec4(I_diffuse+I_ambient, 1.0);  
16 }
```

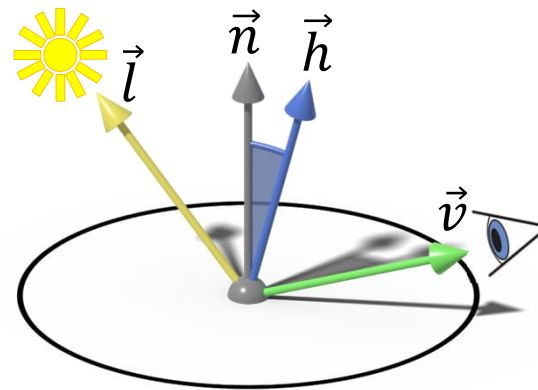


$$\vec{r} = 2 \cdot \vec{n} \cdot (\vec{n} \cdot \vec{l}) - \vec{l}$$
$$I_{\text{specular}} = k_s \cdot L_s \cdot \max(0, \vec{r} \cdot \vec{v})^p$$

<https://bit.ly/2XbPc7s>

Blinn-Phong Illumination Model 1/2

- Alternative simulation of specular reflections
 - Replace mirror direction $\vec{r}$ with Halfway-Vector $\vec{h}$
 - $\vec{h} = |\vec{l} + \vec{v}|$
 - $\vec{h}$ is halfway between $\vec{l}$ and $\vec{v}$
- Equation
- $I_{\text{specular}} = k_s \cdot L_s \cdot \max(0, \vec{h} \cdot \vec{n})^p$



[Blinn, 1977]

Blinn-Phong Illumination Model 2/2

```
1 precision highp float;
2 varying vec3 pos;
3 varying vec3 n;
4 varying vec3 l;
5
6 void main() {
7     vec3 kd = vec3(0.4, 0.4, 0.8);
8     vec3 Ld = vec3(1.0, 1.0, 1.0);
9     vec3 I_diffuse = kd * Ld * max(0.0, dot(n,l));
10
11     vec3 ka = vec3(0.3, 0.3, 0.3);
12     vec3 La = vec3(0.5, 0.5, 0.5);
13     vec3 I_ambient = ka * La;
14
15     vec3 ks = vec3(1.0, 1.0, 1.0);
16     vec3 Ls = vec3(1.0, 1.0, 1.0);
17     float p = 20.0;
18     vec3 v = normalize(-pos);
19     vec3 r = vec3(2) * n * dot(n,l) - l;
20     vec3 I_specular = ks * Ls * pow(max(0.0, dot(r,v)), p);
21
22     gl_FragColor = vec4(I_diffuse+I_ambient+I_specular, 1.0);
23 }
```

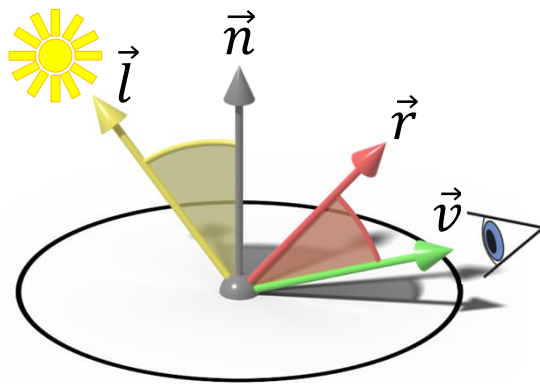


<https://bit.ly/2ICDwBD>

$$\vec{h} = |\vec{l} + \vec{v}|$$
$$I_{\text{specular}} = k_s \cdot L_s \cdot \max(0, \vec{h} \cdot \vec{n})^p$$

Phong Illumination Model

- Additive combination of all factors



$$I_{ambient} = k_a \cdot L_a$$

$$I_{diffuse} = k_d \cdot L_d \cdot \max(0, \vec{n} \cdot \vec{l})$$

$$I_{specular} = k_s \cdot L_s \cdot \max(0, \vec{v} \cdot \vec{r})^p$$

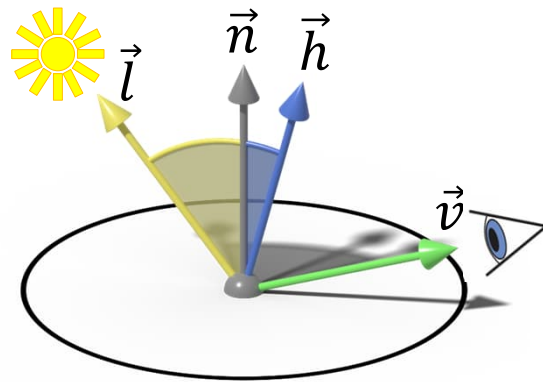
$$I_{Phong}$$



[Phong, 1975]

Phong Illumination Model

- Additive combination of all factors



$$I_{ambient} = k_a \cdot L_a$$

$$I_{diffuse} = k_d \cdot L_d \cdot \max(0, \vec{n} \cdot \vec{l})$$

$$I_{specular} = k_s \cdot L_s \cdot \max(0, \vec{n} \cdot \vec{h})^p$$

$$I_{Blinn-Phong}$$



[Blinn, 1977]



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8.5 Shading

Coloring triangles when using vertex-based illumination



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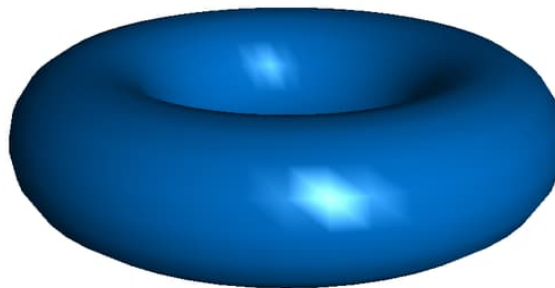
Shading

- Illumination model usually applied in the fragment stage
- Vertex-based illumination also possible to save compute
 - Requires to spread the illumination information of the vertices over the triangle

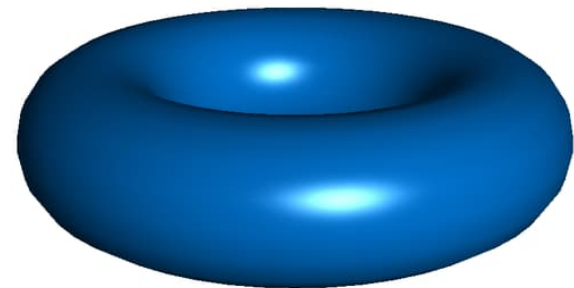
Flat



Gouroud



Phong

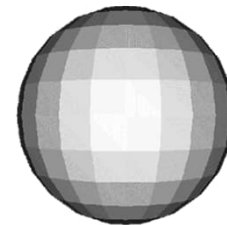
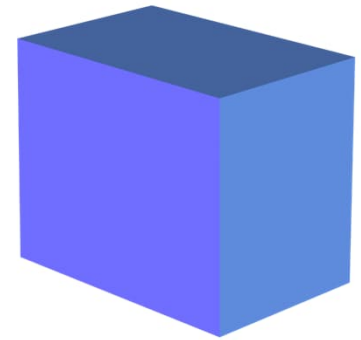


Shading Computation

- Tasks of shading procedures
 - Colorize surfaces based on colors calculated by illumination model
 - Shading is closely linked to the rasterization process and integrated into it
- Distribute lighting and shading
 - Calculate light intensity for surface points
 - Shadow surface sections based on the calculated light intensities
- Basic conflict
 - Illumination model in the camera coordinate system,
shading model in the normalized projection coordinate system

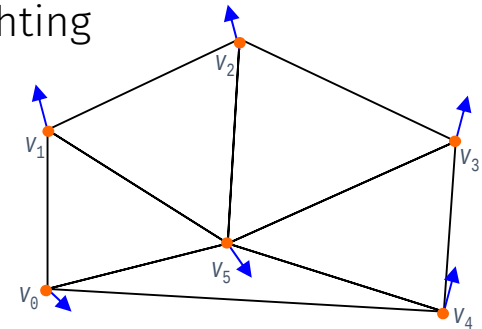
Flat Shading

- Operation
 - Calling the lighting model for a single point of a surface (e.g., polygon center)
 - Shading of the entire polygon in the determined intensity
- Useful if the following assumptions apply
 - Light source is infinitely far away ($\vec{n} \cdot \vec{l}$ regardless of position)
 - Viewer is infinitely far away ($\vec{n} \cdot \vec{v}$ regardless of position)
 - Polygon is a single surface and not part of a surface approximation (e.g., generated by tessellation)
- Otherwise: "faceted" appearance of the objects



Gouraud Shading 1/5

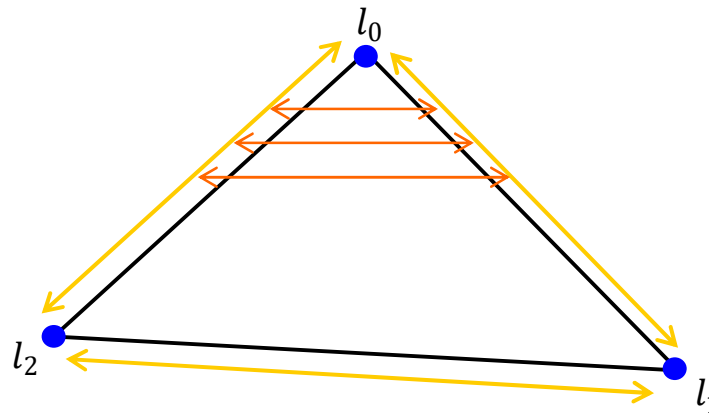
- Shading method for polygons based on the interpolation of intensity values at the polygon vertices
- Requirements
 - Area decomposition in polygons (usually triangles)
 - Normals for polygon corners (vertex normals)
 - Vertex normals are e.g. derivable from an analytical surface description together with the polygonal approximation
 - Vertex normals are e.g. calculated by averaging and weighting the polygon normals of adjacent surfaces



Gouraud Shading 2/5

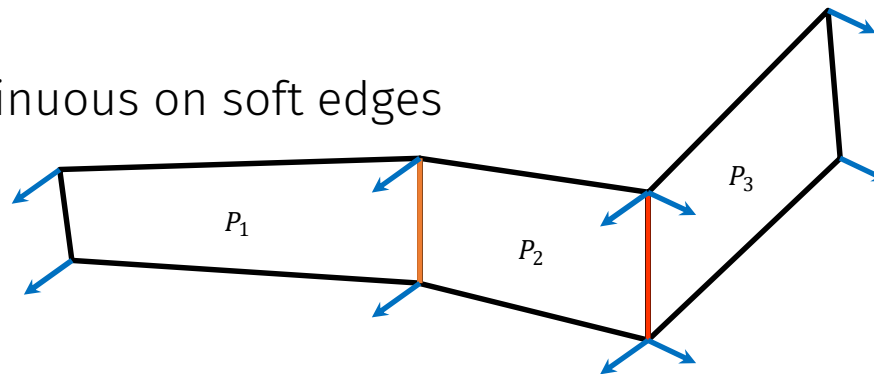
- Proceeding

- Compute intensity values l_i for polygon corners v_i
- Linear interpolation of the intensity values along the edges
- Linear interpolation of the intensity values along the scanlines



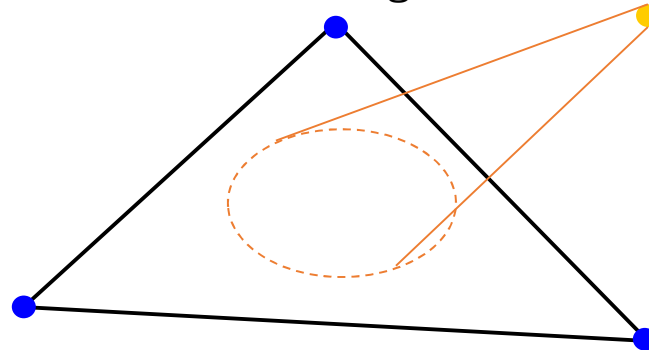
Gouraud Shading 3/5

- Shading at conceptually smooth transitions
 - Geometrically "hard" edges instead of conceptually "soft" edges
 - Gouraud shading can create visually smooth transitions through smooth shading
 - Example: P_1 / P_2 smooth transition, P_2 / P_3 hard transition
 - Necessary: edges on smooth transitions have the same corner normals (shared vertex normals)
 - => Shading process is continuous on soft edges



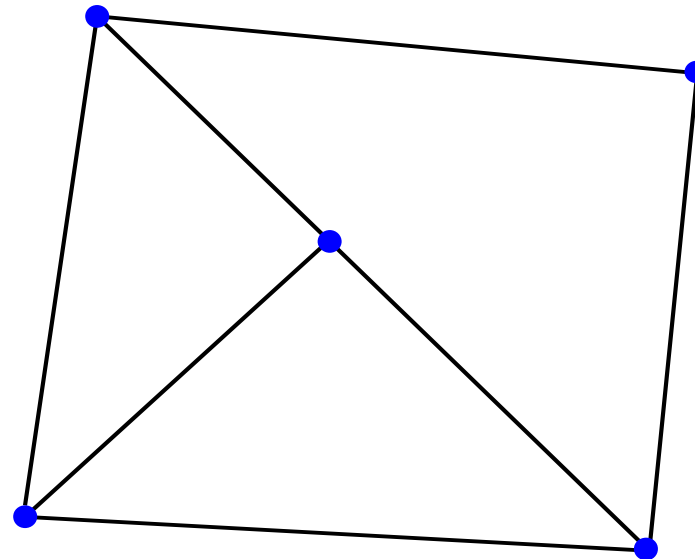
Gouraud Shading 4/5

- interpolation problems
 - The shading inside a triangle results from interpolation
 - Intensity fluctuations in the interior are not taken into account
 - Intensity fluctuations can only be considered by finer tessellation
- Example: Spotlight shines into the interior of a triangle



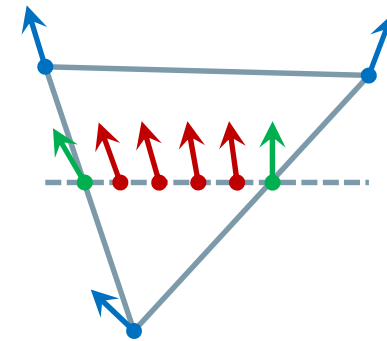
Gouraud Shading 5/5

- Interpolation problems
 - Contiguous polygons that do not fully share the same edges will be shaded differently



Phong Shading 1/2

- Interpolation of the corner normals instead of the corner intensities
- Proceeding
 - Calculation of the corner normals
 - Interpolation of the normals between the corners
 - Interpolation between the endpoints of a scanline
 - Per fragment of a scanline
 - Normalization of the interpolated normal vector
 - Calling the lighting model
 - Coloring of the fragment



Phong Shading 2/2

- Advantages
 - Pixel-based (i.e., image-precise) evaluation of the light sources
 - Highly focused highlights at corners are not interpolated over the edge
 - Highlights inside a triangle can be displayed
- Disadvantage
 - High calculation effort due to renormalization of the interpolated normals



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References

Literature and other sources used in this chapter



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Lesen und Ausprobieren

■ Zum Lesen



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Gouraud, Henri:
Continuous Shading of Curved Surfaces,
IEEE Trans Comput, vol. 20, no. 6 (1971): 623-629.



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■ Zum Ausprobieren

- Simulation von drei ausgewählten Materialien
- Ermittlung der Wertebereiche
- Erzeugen eines Differenzbildes zwischen Phong und Blinn-Phong
- Erweiterung von Phong auf mehrere Lichtquellen
- ...

<https://bit.ly/2XbPc7s>